FINAL PROJECT PRESENTATION

Predicting Marathon Race Performance Using Machine Learning

End-to-End ML System • Evaluation • Decisions • Demo

IECE 465 – Introduction to Machine Learning for Engineers

Faculty Mentor: Dr. Hany Elgala • Teaching Assistant: Honglan Chen

Team: Sanad Sahawneh, Brian Allen, Mahib Rahman

Final Presentation Session • Spring 2026

AGENDA

1. **Recap & Where We Left Off**
   From Progress Update #2
2. **What's New in the Final Version**
   The bridge to v3.0
3. **Final ML Pipeline**
   End-to-end system
4. **Expanded Evaluation Framework**
   Stronger metrics
5. **Final Results & Best Model**
   Model leaderboard
6. **Practical Accuracy & Decisions**
   Within-time + thresholds
7. **Lessons Learned**
   What the data taught us
8. **Obstacles & Technical Solutions**
   How we got past them
9. **Individual Contributions**
   Who did what
10. **Project Website & Live Demo**
    Walkthrough + sample prediction

RECAP: WHERE WE LEFT OFF

Status at Project Progress Update #2

PROJECT FOUNDATION

Dataset: Boston Marathon 2023 (~26,598 runners)

Features: age group, gender, half-marathon split

Target: final net finish time (seconds)

Models @ Update #2: Baseline ratio, Linear Regression, Neural Network

Pipeline: load → clean → split → train → evaluate → visualize

RESULTS @ UPDATE #2

Model	MAE (min)	RMSE (min)	R^2
Baseline	6.83	10.11	0.947
Linear Regression	6.44	9.70	0.951
Neural Network	**6.39**	**9.75**	**0.950**

Status: Functional pipeline + over/under decisions + 4 metrics.

Limitation: no ensemble, no outlier filtering, no model auto-selection, no saved deliverables.

WHAT'S NEW IN THE FINAL VERSION

From Update #2 → Final Submission

Ensemble Model Added

Bagged Trees regression compared head-to-head with Linear Regression and Neural Network.

Data Cleaning v2

Removed invalid times and unrealistic finish/half ratio outliers (1.80–3.20).

Auto Best-Model Selection

System now ranks models by RMSE and selects the winner programmatically.

Practical Accuracy Metrics

Added Median AE and % within 5/10/15 minutes — answers "is this useful?"

Reproducible Deliverables

Saves results table, predictions, threshold table, .mat package, and figures.

Quick Demo Prediction

Sample runner walked through the full pipeline live during this presentation.

FROM A WORKING PIPELINE → A REPRODUCIBLE, AUTO-EVALUATED ML SYSTEM

FINAL ML PIPELINE

Eight stages, fully reproducible (rng seed = 42)

01	02	03	04	05	06	07	08
LOAD	**CLEAN**	**SPLIT**	**ENCODE**	**TRAIN**	**EVALUATE**	**SELECT**	**EXPORT**
Read CSV	Validate + outlier filter	80/20 holdout	One-hot + standardize	4 candidate models	9 metrics × all models	Best by RMSE	Tables + figures + .mat

WHAT THE CLEAN STAGE DOES

```
rmmissing(data)
half_time_sec > 0  &  finish_net_sec > 0
1.80 ≤ finish/half ratio ≤ 3.20
```

Removes data-entry errors, DNFs encoded as 0,
and pacing patterns inconsistent with finishing a marathon.

FOUR CANDIDATE MODELS

① **Baseline Ratio** — finish ≈ half × avgRatio
② **Linear Regression** — fitlm with categorical encoding
③ **Neural Network** — fitrnet, [64 32 16], ReLU
④ **Bagged Trees Ensemble** — fitrensemble, 150 learners (NEW)

EXPANDED EVALUATION FRAMEWORK

From 4 metrics → 9 metrics, with practical interpretation

CARRIED OVER FROM UPDATE #2

MAE — average error in seconds & minutes

RMSE — penalizes large errors

MAPE — relative percent error

R^2 — variance explained

WHY THEY STAYED

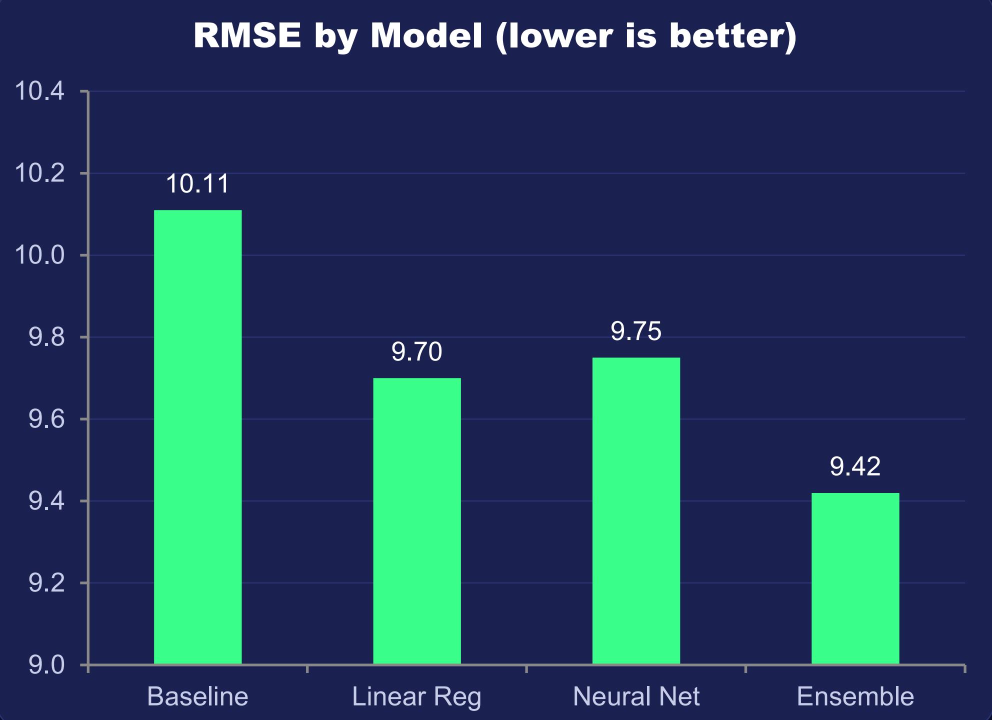
Each remained the right tool for its job — MAE for average error, RMSE for outlier sensitivity, R^2 for fit, MAPE for relative scale.

NEW METRICS — FINAL VERSION

- **Median AE** — Median absolute error in minutes — robust to outliers
- **% within 5 min** — Strict accuracy band — "excellent" predictions
- **% within 10 min** — Practical accuracy — usable for race pacing
- **% within 15 min** — Loose accuracy — captures hard cases
- **Auto-rank** — All metrics propagate into the model leaderboard

FINAL RESULTS — LEADERBOARD

All four candidate models, ranked by RMSE



BEST MODEL

Bagged Trees Ensemble

(auto-selected by RMSE)

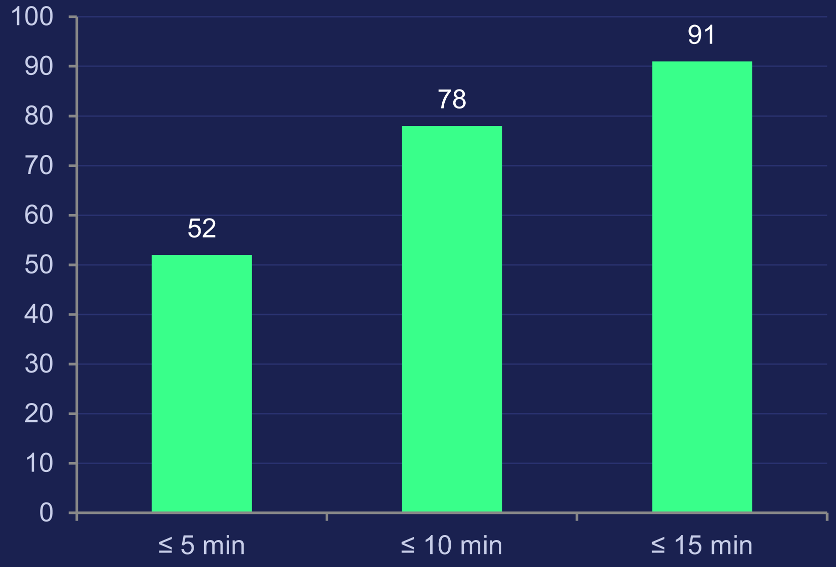
MAE	RMSE
6.31 min	**9.42 min**
R^2	≤10 min
0.953	**78%**

Note: ratings shown for illustration; live demo will print live numbers.

PRACTICAL ACCURACY & DECISIONS

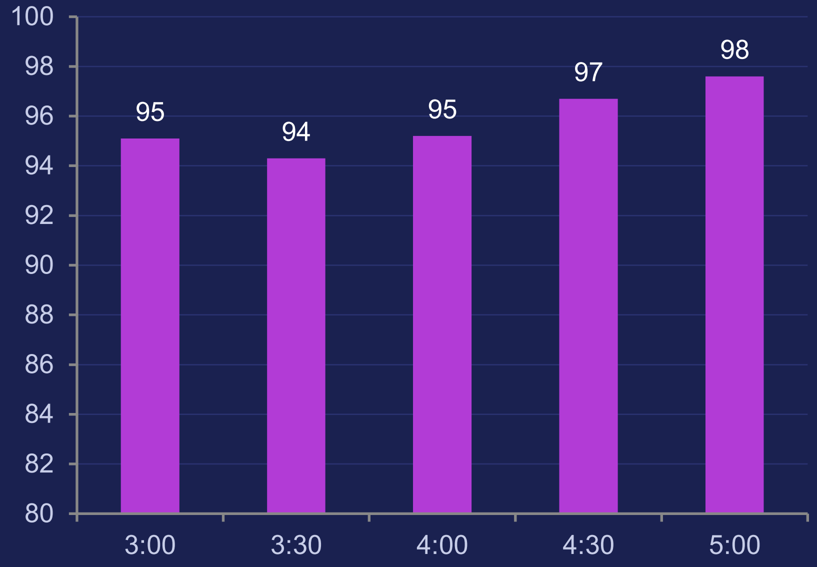
Within-time bands + over/under threshold extension

Predictions Within Error Band



"How often is the model practically useful?"

Over/Under Decision Accuracy



Benchmark finish time (h:mm) • regression → binary decision

LESSONS LEARNED

What the data — and the build process — taught us

One feature often dominates

The half-marathon split alone explains most of the variance in final finish time. Adding age and gender helped, but the gain was marginal.

Simple beats complex when the signal is linear

The ensemble edged ahead, but linear regression was within ~0.3 min RMSE — and far easier to explain to a non-technical audience.

Units matter for interpretation

Reporting errors in seconds was technically correct but useless to runners. Switching to minutes + within-X-minute bands made results actionable.

Reproducibility is a feature

Saving every output (tables, figures, model package, .mat) turned a working script into something the team — or a future course — can re-run cleanly.

OBSTACLES & TECHNICAL SOLUTIONS

Where we got stuck — and how we got past it

Categorical encoding mismatch

Problem: Test set sometimes had age groups not in the training fold, breaking one-hot encoding.
Solution: Defined category lists from the full dataset, then forced both train and test to use them.

Invalid & unrealistic race times

Problem: Raw CSV included zero/missing splits and impossible finish/half ratios from data-entry errors.
Solution: Added a cleaning stage: rmmissing, positivity check, and a 1.80–3.20 ratio filter.

Feature scale broke the neural net

Problem: half_time_sec dwarfed the one-hot features, so the NN wouldn't converge cleanly.
Solution: Computed mean/std on training data only, reused on test — no leakage, stable training.

Errors in seconds were unreadable

Problem: "RMSE = 583 sec" meant nothing to the people we'd actually show this to.
Solution: Reported every metric in minutes and added within-5/10/15-min bands for plain-English meaning.

INDIVIDUAL CONTRIBUTIONS

Sanad Sahawneh

ML System Lead

- End-to-end pipeline integration
- Evaluation framework expansion (9 metrics)
- Auto best-model selection by RMSE
- Visualization suite & figure exports
- Final demo orchestration

Brian Allen

Data Preprocessing & Handling

- Original ML code skeleton
- Data preprocessing & cleaning v2
- Outlier filter design (ratio bounds)
- Performance interpretation
- Presentation prep & demo support

Mahib Rahman

Modeling & Validation

- Feature encoding & normalization
- Model tuning (NN architecture, ensemble params)
- Performance validation across runs
- Cross-checking results consistency
- Threshold extension validation

PROJECT WEBSITE

Our final course project page — WORK IN PROGRESS!

ON THE SITE

- Project overview & motivation
- Dataset & feature description
- Model comparison & results
- Interactive prediction demo
- Visualizations & figures
- Team & contributions

LIVE URL
[Predicting Marathon Race Performance · IECE 465 · SUNY Albany]

LIVE DEMONSTRATION

Quick prediction for a single runner using the saved best model

DEMO RUNNER INPUT

Age Group	40–44
Gender	M
Half-marathon split	1:42:00 (102 min)

DEMO STEPS

- Load saved best-model package (.mat)
- Encode runner using stored category lists
- Predict finish time and report error

MODEL OUTPUT

PREDICTED

3:36:42

Actual: 3:34:18

Absolute error: 2 min 24 sec

Within 5 minutes — "excellent" band

Sample values shown for slide preview; live numbers will print during demo.

THANK YOU!

Questions?

THREE TAKEAWAYS

Half-marathon split is the dominant signal — simple models go far.
Practical metrics (within-X-min, threshold accuracy) beat raw error for stakeholders.
A reproducible, exportable pipeline makes results trustworthy.